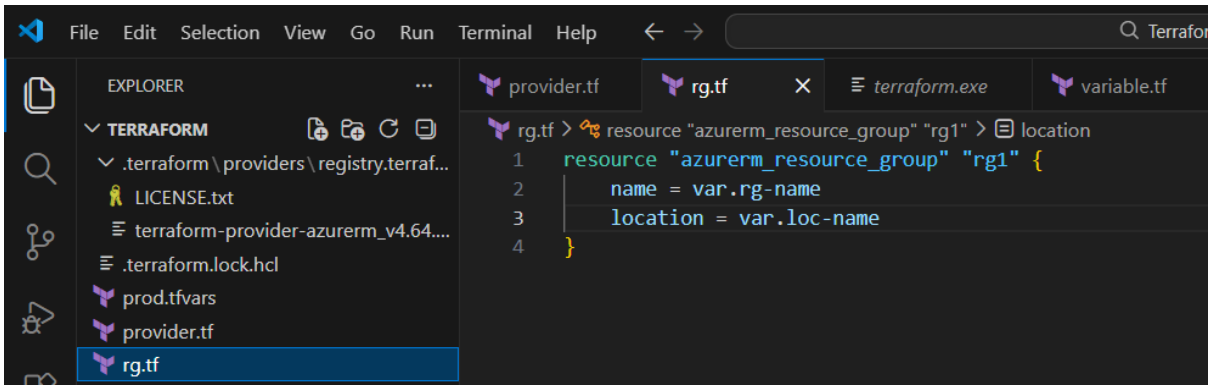
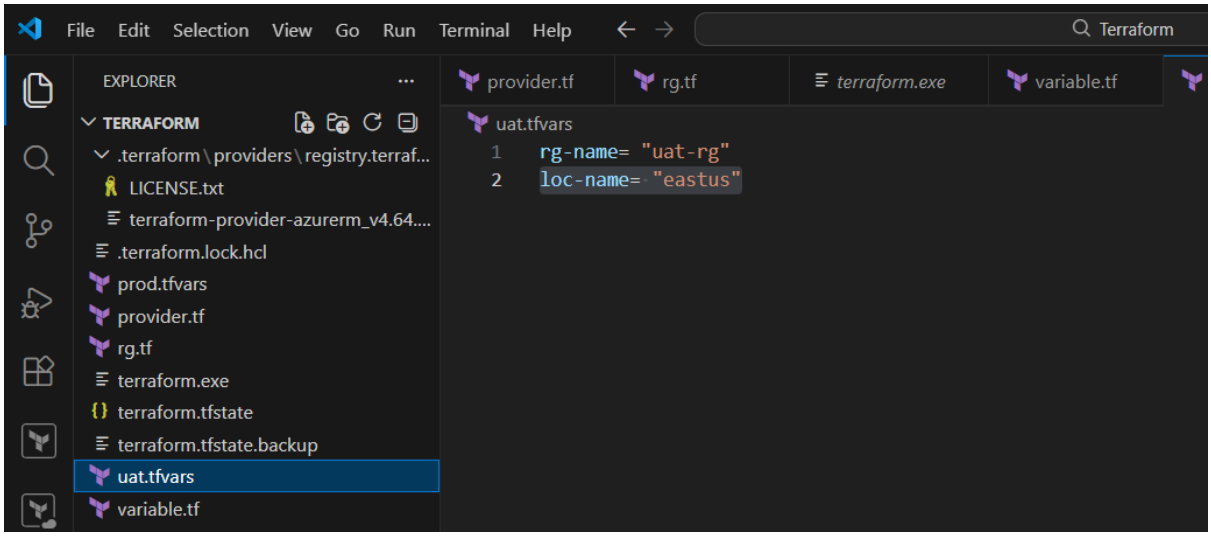
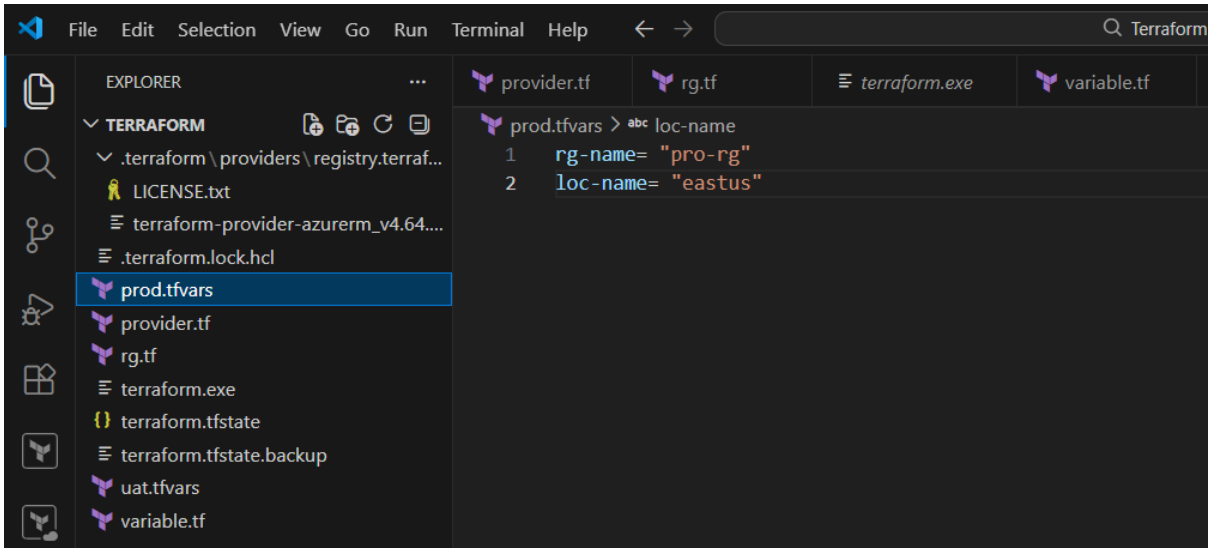
Workspace

✓ To maintain multiple Statefiles in multiple Environments.



- Created Two different environment files Prod and uat to access the values based on the requirements.






```
PS C:\Terraform> terraform plan
var.loc-name
  Enter a value: |
```

- It Asking to Enter the Values Because We Given the 2 tfvars files.
- So we have to provide the path of the file to create the resources based on that values in it.
  ✓ Terraform plan terraform plan --var-file .\uat.tfvars

```
PS C:\Terraform> terraform plan --var-file .\uat.tfvars

Terraform used the selected providers to generate the following execution plan. Resource actions are
  + create

Terraform will perform the following actions:

  # azurerm_resource_group.rg1 will be created
  + resource "azurerm_resource_group" "rg1" {
      + id       = (known after apply)
      + location = "eastus"
      + name     = "uat-rg"
    }

Plan: 1 to add, 0 to change, 0 to destroy.
```

```
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these a
PS C:\Terraform> terraform apply --var-file .\uat.tfvars --auto-approve

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  # azurerm_resource_group.rg1 will be created
  + resource "azurerm_resource_group" "rg1" {
      + id       = (known after apply)
      + location = "eastus"
      + name     = "uat-rg"
    }

Plan: 1 to add, 0 to change, 0 to destroy.
azurerm_resource_group.rg1: Creating...
azurerm_resource_group.rg1: Still creating... [00m10s elapsed]
azurerm_resource_group.rg1: Still creating... [00m20s elapsed]
azurerm_resource_group.rg1: Creation complete after 30s [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e168

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Terraform>
```

- While Creating another vars –prod.tfvars it may delete the uat-rg and it will create the prod-rg.
- To avoid these we use **Workspace**.

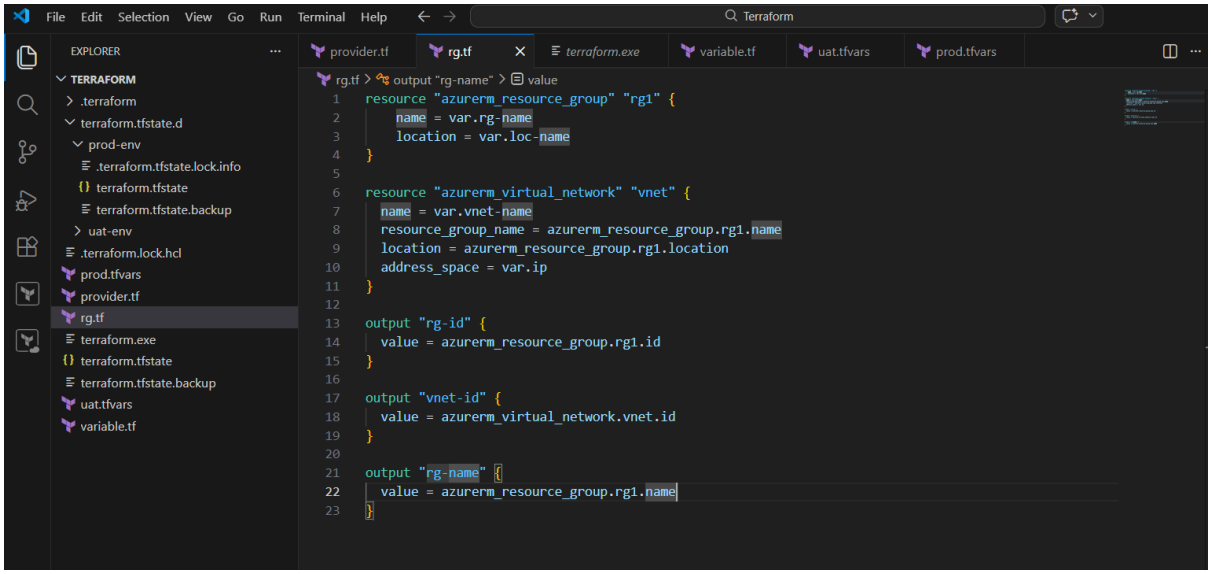
```
PS C:\Terraform> terraform workspace
Usage: terraform [global options] workspace

  new, list, show, select and delete Terraform workspaces.

Subcommands:
    delete    Delete a workspace
    list      List Workspaces
    new       Create a new workspace
    select    Select a workspace
    show      Show the name of the current workspace
PS C:\Terraform> terraform workspace show
default
PS C:\Terraform> terraform workspace new uat-env
Created and switched to workspace "uat-env"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Terraform> terraform workspace new prod-env
Created and switched to workspace "prod-env"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Terraform> terraform workspace show
prod-env
PS C:\Terraform> terraform workspace list
  default
* prod-env
  uat-env
```

✓ To Switch from one Environment to Another.

```
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Terraform> terraform workspace select prod-env
PS C:\Terraform> terraform workspace show
prod-env
PS C:\Terraform>
```

✓ Every Folder as its own Statefile.

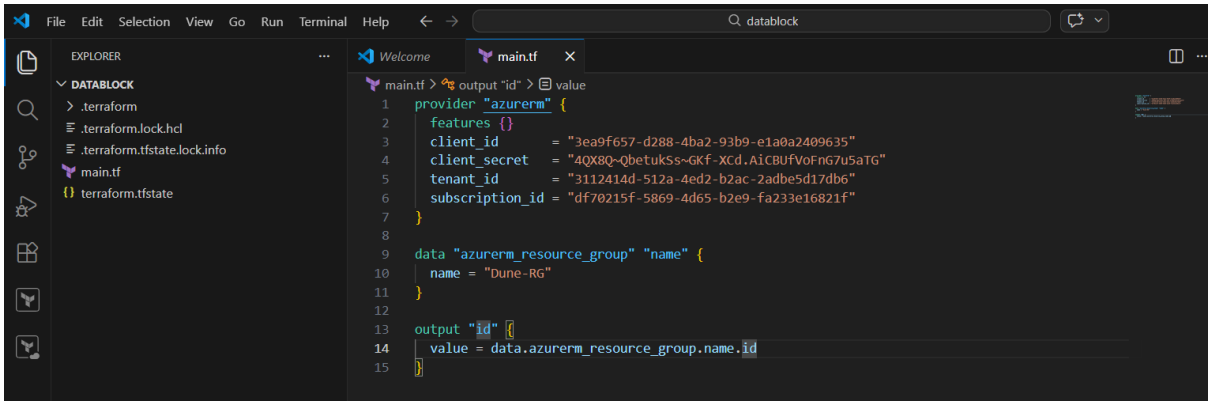
- To get ids or names as output, we use **Output Block.**




```
Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
PS C:\Terraform> terraform apply --var-file .\prod.tfvars --auto-approve
azurerm_resource_group.rg1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups
azurerm_virtual_network.vnet: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGrou
virtualNetworks/vnet01]

Changes to Outputs:
  + rg-id   = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/uat-rg"
  + rg-name = "uat-rg"
  + vnet-id = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/uat-rg/providers/Microsoft.Network/v

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructu

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

rg-id = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/uat-rg"
rg-name = "uat-rg"
vnet-id = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/uat-rg/providers/Microsoft.Network/virtu
PS C:\Terraform>
```

Data Block

- Data Block is used to read the existing infrastructure.



- To get the id of the existing Resource Group.

```
Plan: 0 to add, 0 to change, 1 to destroy.

Changes to Outputs:
  + id = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/Dune-RG"
azurerm_resource_group.rg: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/databl
azurerm_resource_group.rg: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/
azurerm_resource_group.rg: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/
azurerm_resource_group.rg: Destruction complete after 27s

Apply complete! Resources: 0 added, 0 changed, 1 destroyed.

Outputs:

id = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/Dune-RG"
PS C:\provisionerterraform\datablock>
```